

5.8 GHz 2x2 MIMO FMCW RADAR ARRAY - ZYF300CA-P PTFE

FABRICATION NOTES

- 1. MATERIAL: ZYF300CA-P PTFE, dielectric thickness 0.76 mm +/-10%.
Design permittivity $\epsilon_r = 3.0$ (microstrip design value), $\tan \delta = 0.0018$.
No substitution without approval: the patch dimensions are tuned to this permittivity.
- 2. LAYERS: 2 copper layers. Layer 1 = antenna and feed network. Layer 2 = SOLID ground.
The ground plane must be continuous. Do NOT add thieving, hatching or venting to layer 2.
- 3. COPPER: 1 oz (35 μm) finished on both layers.
- 4. FINISH: IMMERSION SILVER per IPC-4553. DO NOT SUBSTITUTE ENIG.
At 5.8 GHz the current reaches only 0.87 μm into the metal, so ENIG's 3 μm of nickel would carry all of it, at 2.7 times the resistance of copper -- 4.3 per cent of detection range.
Silver tarnish is a semiconductor and carries no RF current: a blackened board measures the same as a fresh one.
- 5. SOLDER MASK: layer 1 has deliberate openings over the radiating elements and the four end-launch transitions - mask there would detune the patches and the launches.
Mask on layer 2 as normal. Green LPI.
- 6. SILKSCREEN: white, layer 1 and layer 2 as supplied. Do not print silkscreen on bare copper.
- 7. IMPEDANCE: single-ended 50 ohm +/-10% on the 1.854 mm traces, referenced to layer 2.
Impedance control is requested; adjust trace width if your stack requires it, and report the width used. Do NOT alter the patch or transformer geometry.
- 8. ETCH TOLERANCE: +/-0.02 mm on conductor width. Minimum feature on this board is 0.341 mm (the quarter-wave transformers) - confirm you can hold it before starting.
- 9. HOLES: 14 x 0.40 mm plated stitching vias; 4 x 3.20 mm plated mounting holes.
- 10. OUTLINE: routed, +/-0.15 mm. Copper reaches the board edge at the four launch sites by design.
- 11. NO panelisation tabs across the antenna areas. V-score not permitted.
- 12. ELECTRICAL TEST: 100% netlist test.
- 13. The mask openings in note 5 expose adjacent copper of different nets.
That is intended on an antenna board and is the only rule waived in the DRC report; everything else passes at error AND warning severity.

STACKUP

+-----+ layer 1 35 μm Cu, immersion silver, mask open over antennas			
ZYF300CA-P PTFE	0.760 mm	ϵ_r 3.0, $\tan \delta$ 0.0018	
+-----+ layer 2 35 μm Cu, SOLID ground, masked			
finished thickness approx 0.83 mm			

DRILL TABLE

SIZE	QTY	PLATED	NOTE
0.40 mm	14	yes	ground stitching
3.20 mm	4	yes	M3 mounting, bonded to ground

ELEMENT POSITIONS (board origin at lower-left, mm)

these are the phase centres to enter into the beamforming code

TX1	RHCP	x	22.9990	y	34.1950
TX2	RHCP	x	48.8432	y	34.1950 (mirrored)

CRITICAL RF DIMENSIONS (do not modify)

patch	14.1135 x 14.1135 mm
element pitch	25.8442 mm (half wave in air at 5.800 GHz)
50 ohm trace width	1.8535 mm
transformer	0.3414 mm wide x 8.5422 mm long (110 ohm)
hybrid arms	8.8450 mm (vertical) / 8.9982 mm (horizontal)
launch ground gap	0.9699 mm each side

OUTLINE AND DRILL PLAN 71.34198 x 47.237 mm
(dimensions in mm, origin at lower-left)

